

SYSTEMS of LINEAR EQUATIONS

SKILLS PRACTICE & WORD PROBLEMS

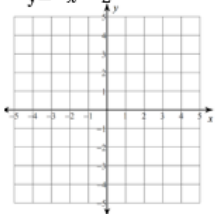
PRACTICE WORKSHEETS

Solving Systems of Linear Equations by Graphing

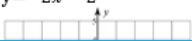
Skills Practice

Solve the system of linear equations by graphing.

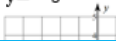
1. $y = 4x + 3$
 $y = -x - 2$



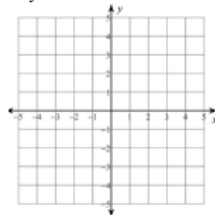
2. $y = x + 4$
 $y = -2x - 2$



3. $y = 6x - 3$
 $y = -3$



4. $y = 3x + 4$
 $y = -x - 4$



Problem Solving

There are five students wearing g
are three more boys than girls we
7. Write a system of equations to
number of boys and girls wear

8. Solve the system of linear equ
graphing.

- ✓ Solving Systems of Linear Equations by Graphing
- ✓ Solving Systems of Linear Equations by Substitution
- ✓ Solving Systems of Linear Equations by Elimination

Solving Systems of Linear Equations by Elimination

Skills Practice

Solve the system of linear equations by elimination. Check your solution.

3. $x + 3y = -2$
 $x - 3y = 16$

6. $3x - 4y = 33$
 $4x + 3y = 19$

Solving Systems of Linear Equations by Substitution

Skills Practice

Solve the system of linear equation by substitution. Check your solution.

1. $2x + 3y = 0$
 $8x + 9y = 18$

2. $-x + 5y = 28$
 $x + 3y = 20$

3. $-3x + y = 2$
 $-x + y - 4 = 0$

4. $y = 3x - 4$
 $5x - 2y = 10$

5. $y = 3x + 1$
 $y = x + 3$

6. $\frac{3}{4}x - 5y = 7$
 $x = -4y + 12$

Problem Solving

7. The length of a basketball court is twice its width. The perimeter is 150 feet. Find the length and width of the basketball court.

8. Errol cut a 12 foot board into two pieces. The long piece is twice as long as the short one. Write and solve a system of linear equations to find the length of the short piece?

Justin sells brownies for \$2 each and cupcakes for \$3 each. Justin sells a total of 100 brownies and cupcakes for \$240. Write and solve a system of linear equations to find the number of brownies and the number of cupcakes Justin sold.



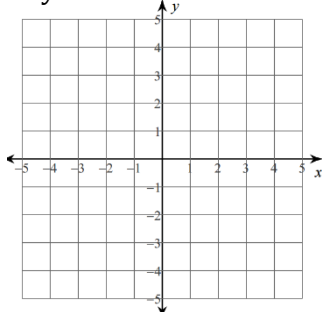
Solving Systems of Linear Equations by Graphing

Skills Practice

Solve the system of linear equations by graphing.

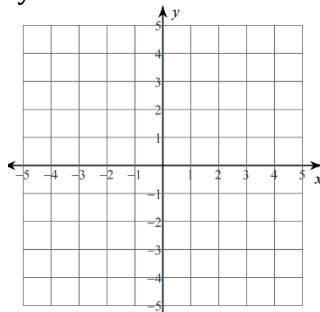
1. $y = 4x + 3$

$y = -x - 2$



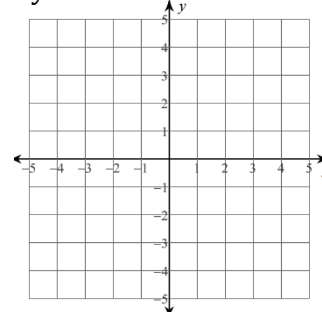
2. $y = x + 4$

$y = -2x - 2$



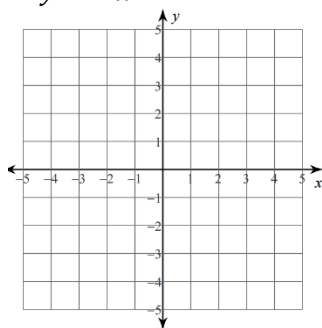
3. $y = 6x - 3$

$y = -3$



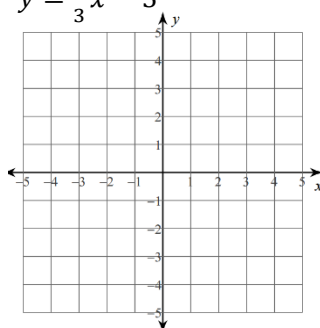
4. $y = 3x + 4$

$y = -x - 4$



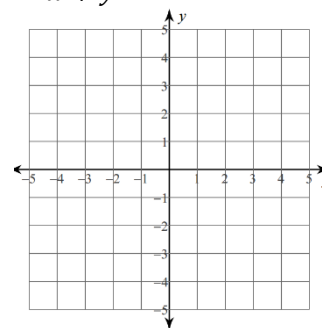
5. $y = -x + 1$

$y = \frac{1}{3}x - 3$



6. $-3x + y = 1$

$2x + y = -4$

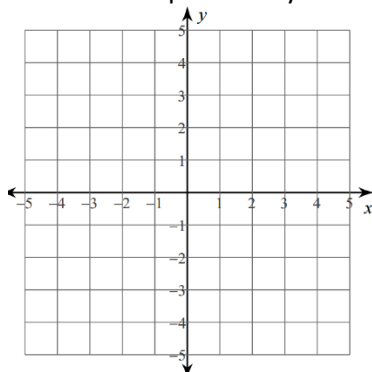


Problem Solving

There are five students wearing glasses. There are three more boys than girls wearing glasses.

7. Write a system of equations to find the number of boys and girls wearing glasses.

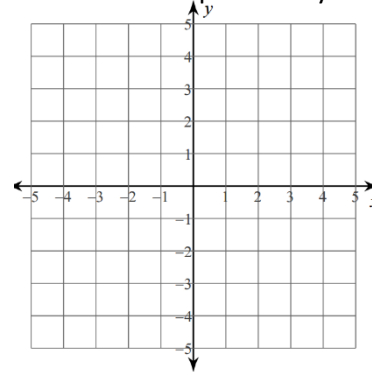
8. Solve the system of linear equations by graphing.



Laura bought a notebook and pencil for \$4.50. The notebook cost twice the amount of the pencil.

9. Write a system of equations to find the cost of the notebook and pencil.

10. Solve the system of linear equations by graphing.



Solving Systems of Linear Equations by Substitution

Skills Practice

Solve the system of linear equation by substitution. Check your solution.

1. $2x + 3y = 0$
 $8x + 9y = 18$

2. $-x + 5y = 28$
 $x + 3y = 20$

3. $-3x + y = 2$
 $-x + y - 4 = 0$

4. $y = 3x - 4$
 $5x - 2y = 10$

5. $y = 3x + 1$
 $y = x + 3$

6. $\frac{3}{4}x - 5y = 7$
 $x = -4y + 12$

Problem Solving

7. The length of a basketball court is twice its width. The perimeter is 150 feet. Find the length and width of the basketball court.

8. Errol cut a 12 foot board into two pieces. The long piece is twice as long as the short one. Write and solve a system of linear equations to find the length of the short piece?

9. Deborah spent \$70 on food and clothes. She spent \$26 more on clothes than on food. Write and solve a system of equations to find how much Deborah spent on each.

10. Gertrude won the jump rope contest four more times than Elana. They have won a combined total of 18 jump rope contests. Write and solve a system of equations to find the number of contests Gertrude won and the number of contests Elana won.

Solving Systems of Linear Equations by Elimination

Skills Practice

Solve the system of linear equations by elimination. Check your solution.

1. $x + 2y = 4$
 $-x - y = 2$

2. $2x + y = 3$
 $x - y = 3$

3. $x + 3y = -2$
 $x - 3y = 16$

4. $-5x + 2y = 13$
 $5x + y = -1$

5. $3x - y = 14$
 $-3x + 4y = 16$

6. $3x - 4y = 33$
 $4x + 3y = 19$

Problem Solving

7. You buy 5 pairs of jeans and 10 t-shirts for \$140. Your friend buys 3 pairs of jeans and 5 t-shirts for \$79. Write and solve a system of linear equations to find the cost of each t-shirt.

8. Justin sells brownies for \$2 each and cupcakes for \$3 each. Justin sells a total of 100 brownies and cupcakes for \$240. Write and solve a system of linear equations to find the number of brownies and the number of cupcakes Justin sold.

9. The seventh and eighth grade students collected canned goods to donate to a local charity. Together they collected 204 canned goods. The eighth grade collected 3 times as many canned goods as the seventh grade. Write and solve a system of linear equations to find how many canned goods the eighth grade students collected.

10. Marcelle uses a total of 6 plates to add 110 pounds to a weight lifting bar. He uses 25-pound plates and 5-pound plates. Write and solve a system of equations to find the number of 25-pound plates and the number of 5-pound plates he uses.

Systems of Linear Equations Practice Worksheets - ANSWER KEY

Solving Systems of Linear Equations by Graphing		
1. $(-1,-1)$	2. $(-2,2)$	3. $(0,-3)$
4. $(-2,-2)$	5. $(3,-2)$	6. $(-1,-2)$
7. $x + y = 5$ $3 + x = y$ 8. $(1,4)$; 4 boys, 1 girl		9. $x + y = 4.50$ $y = 2x$ 10. $(1.5,3)$; notebook \$3, pencil \$1.50

Systems of Linear Equations Practice Worksheets - ANSWER KEY

Solving Systems of Linear Equations by Substitution		
1. (9,-6)	2. (2,6)	3. (1,5)
4. (-2,-10)	5. (1,4)	6. $(11, \frac{1}{4})$
7. $x = 2y$ $2x + 2y = 150$ 50 ft by 25 ft		8. $x + y = 12$ $2x = y$ 4ft
9. $x + y = 70$ $y - x = 26$ clothes: \$48, food: \$22		10. $x + y = 18$ $x = y + 4$ Gertrude: 11, Elana: 7

Solving Systems of Linear Equations by Elimination		
1. (-8,6)	2. (2,-1)	3. (7,-3)
4. (-1,4)	5. (8,10)	6. (7,-3)
7. $5x + 10y = 140$ $3x + 5y = 79$ \$5 each		8. $x + y = 100$ $2x + 3y = 240$ 60 brownies, 40 cupcakes
9. $x + y = 204$ $y = 3x$ 153 canned goods		10. $x + y = 6$ $25x + 5y = 110$ four 25-pound plates, two 5-pound plates

Thank you SO MUCH for purchasing this product!

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PROBLEM SOLVING GRAPHIC ORGANIZERS

PYTHAGOREAN THEOREM

diagonal of a television measures inches. If the width of a 27-inch is inches, state its height to the nearest inch.

A ladder leaning against a building is used to reach a window that is 18 feet above the ground. The bottom of the ladder is 8.75 feet from the building. How long is the ladder? Round to the nearest foot.

✓ Pythagorean Theorem
✓ Converse of Pythagorean Theorem
✓ Solve a Right Triangle
✓ Pythagorean Theorem in 3-Dimensions

Exceeding the CORE

MULTI-STEP EQUATIONS
ERROR ANALYSIS

Error Analysis – MULTI-STEP EQUATIONS

Read the word problem. Look at the student work and solution. Identify the error and describe it. Solve the problem. Share a strategy this student could use to prevent the same error in the future.

Universal Shipping charges \$14 plus \$2 a pound to ship an overnight package. Send-it Quick charges \$1.50 a pound to ship an overnight package. For what weight is the charge the same for the two companies?

Incorrect Work/Solution

$14 + 2x = 20 + 1.50x$
 $-2x$
 $14 = -18 + 1.50x$
 $+18$
 $32 = 1.50x$
 $\div 1.50$
 $21.33 = x$

Both companies charge the same price when the package weighs 21.33 pounds.

✓ Equations with Variables on Each Side
✓ Equations with Rational Numbers
✓ Multi-Step Equations

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